Diabetes Risk Factors

in the Pima Indian Population

A statistical analysis using descriptive statistics, inferential testing, and regression modelling



Unit 12 · MSc Artificial Intelligence · University of Essex

Dataset: Pima Indian Diabetes Database (Kaggle)
Smith et al. (1988); Khare, A.D. (2022)

Presentation Structure

1

Introduction & Dataset

768 Pima Indian women · 9 variables · diabetes outcomes

2

Data Exploration

Distributions, descriptive statistics, outliers

3

Diabetes Rates by Subgroup

BMI, age, and pregnancy categories

4

Inferential Statistics

Hypothesis tests, correlation, and regression models

5

Model Evaluation

Logistic regression performance & interaction effects

6

Conclusions & Recommendations

Findings, limitations, and six recommendations



Section 1

Introduction & Dataset



Dataset Overview: Pima Indian Diabetes Database

Study Population

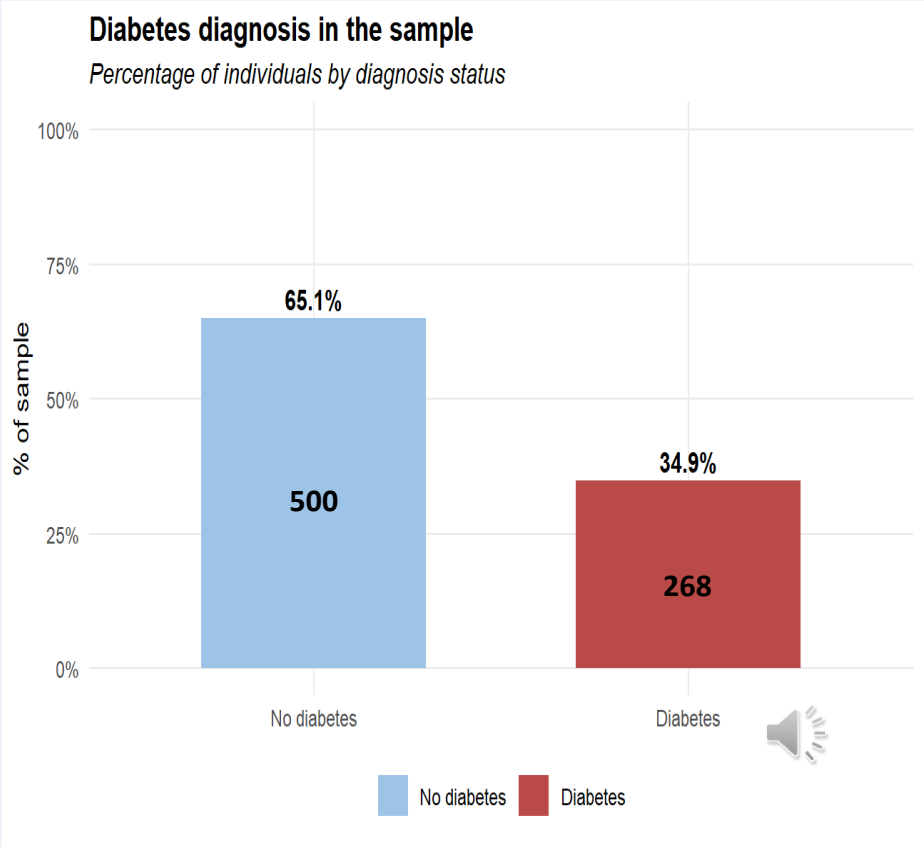
768 Pima Indian women aged 21–81
from the Phoenix, Arizona area

34.9% diabetes prevalence: 3–4× the global average (~10%)

Source: National Institute of Diabetes and Digestive and Kidney Diseases (Smith et al., 1988; Khare, 2022; International Diabetes Federation, 2021)

Variables

- Pregnancies · Glucose · Blood Pressure
- Skin Thickness · Insulin · BMI
- Diabetes Pedigree Function · Age
- Outcome (diabetes diagnosis: 0/1)

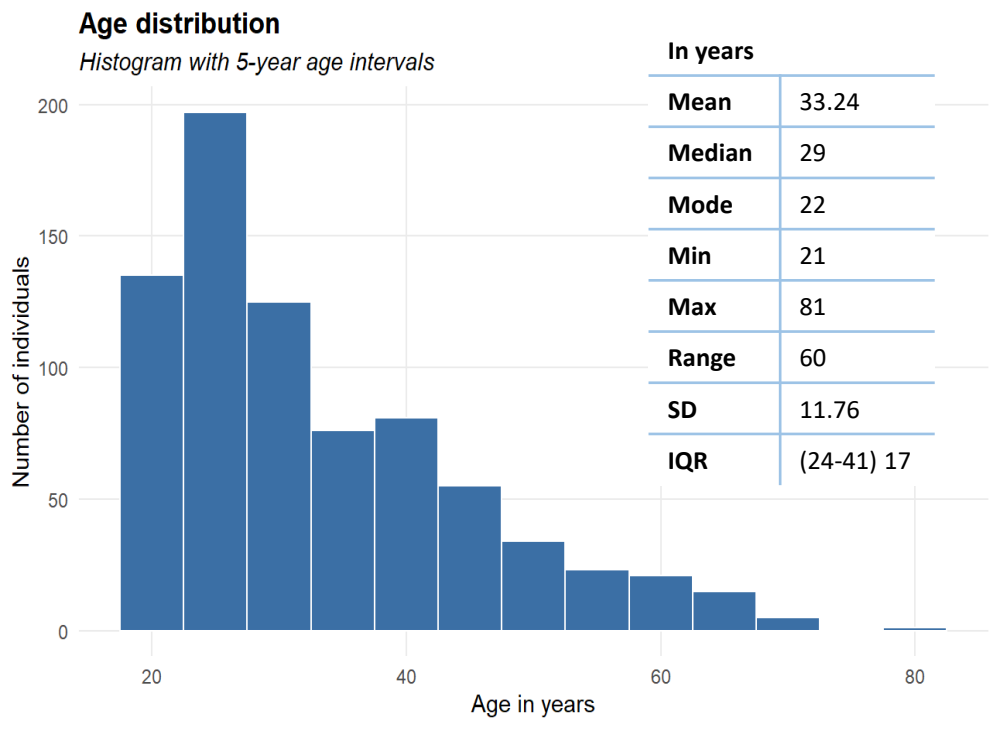


Section 2

Data Exploration & Descriptive Statistics



Age Distribution



In years	
Mean	33.24
Median	29
Mode	22
Min	21
Max	81
Range	60
SD	11.76
IQR	(24-41) 17

Fig 1. Age histogram (5-year intervals)

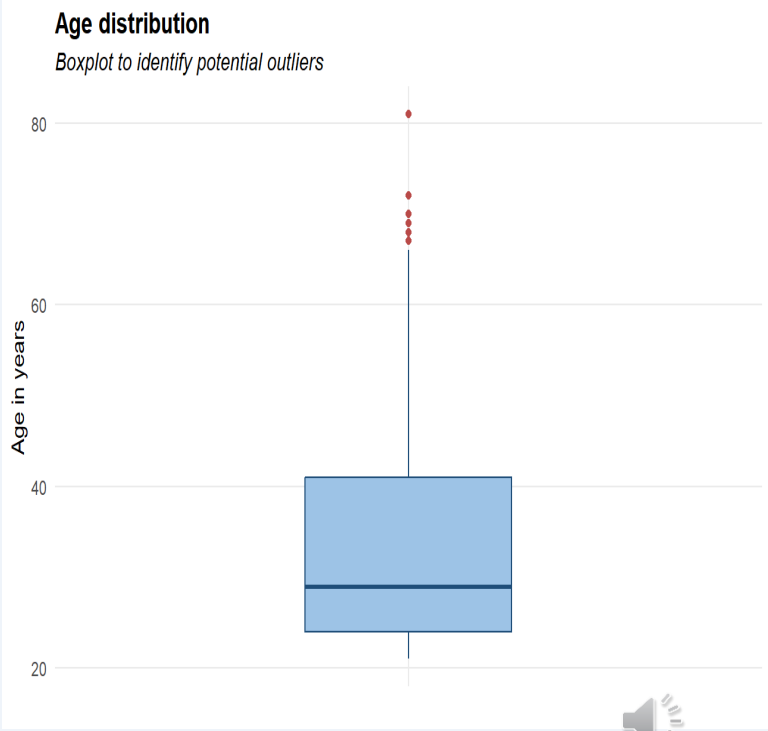


Fig 2. Age boxplot (outliers flagged)

Median age: 29 years · IQR: 24–41 years

Sample is young-skewed, right-tailed distribution

Upper outliers: individuals above the 66.5 fence retained (biologically plausible)

Glucose & Blood Pressure Distributions

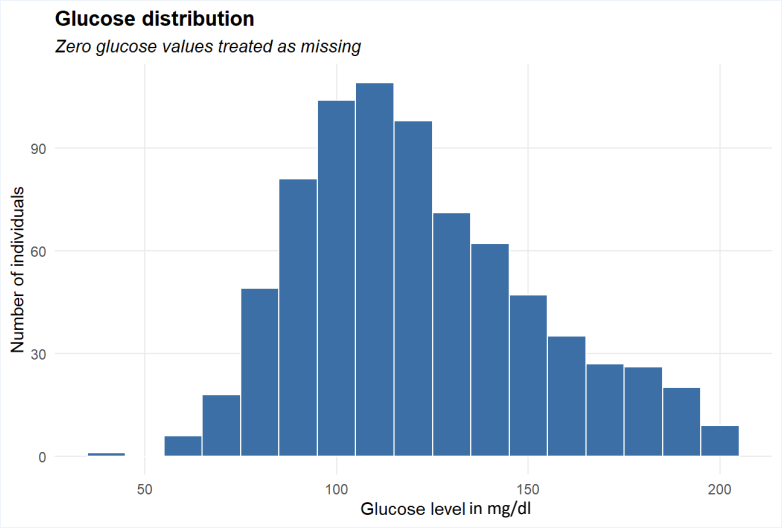
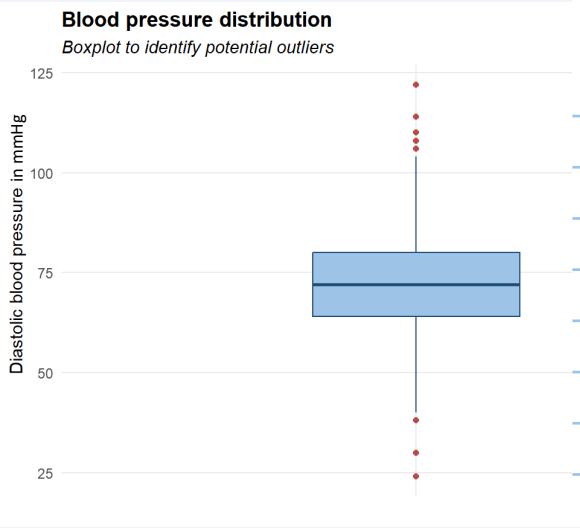


Fig 3. Glucose distribution (zero values = missing)

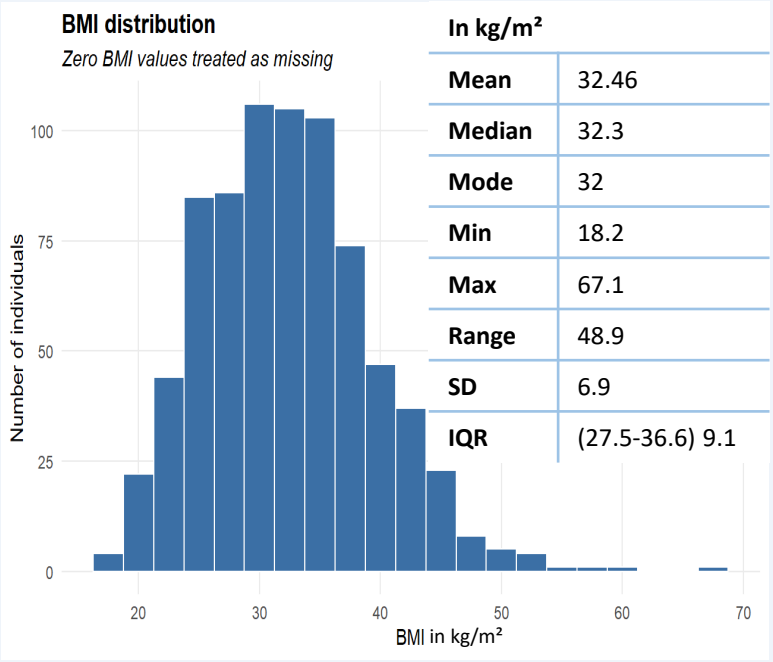


	Glucose in mg/dl	Blood Pressure in mmHg
Mean	121.7	72.4
Median	117	72
Mode	100	70
Min	44	24
Max	199	122
Range	155	98
SD	30.5	12.4
IQR	(99-141) 42	(64-80) 16

Fig 4. Blood pressure distribution (zero = missing)

Zero values in glucose, blood pressure, BMI, skin thickness, and insulin
treated as physiologically implausible missing data, excluded from regression sample
Regression n reduced from 768 → 392 (complete-case analysis)

BMI Distribution



In kg/m²	
Mean	32.46
Median	32.3
Mode	32
Min	18.2
Max	67.1
Range	48.9
SD	6.9
IQR	(27.5-36.6) 9.1

Fig 5. BMI histogram (zero values removed)

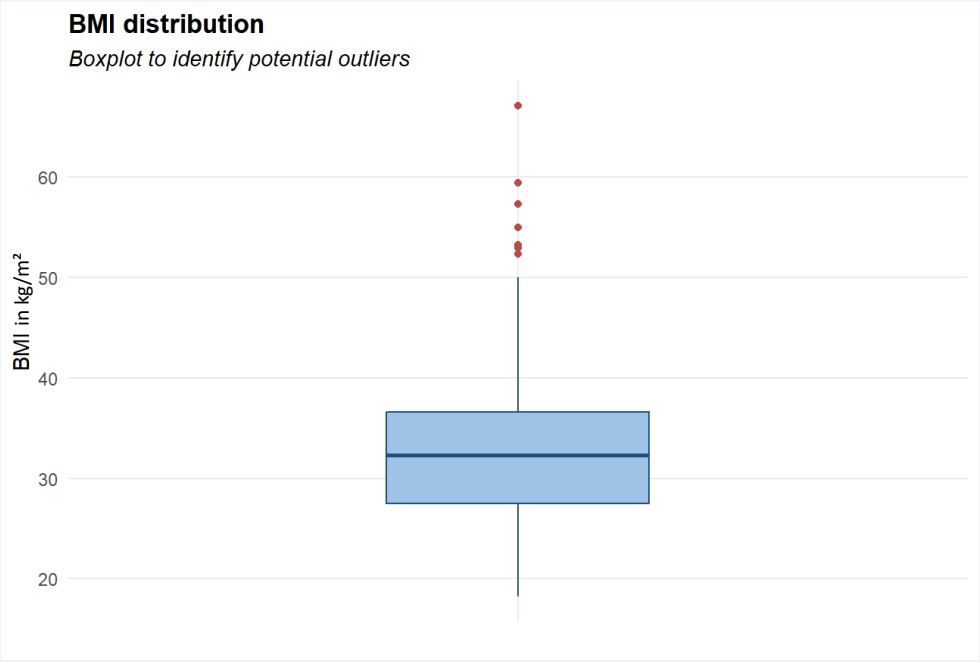
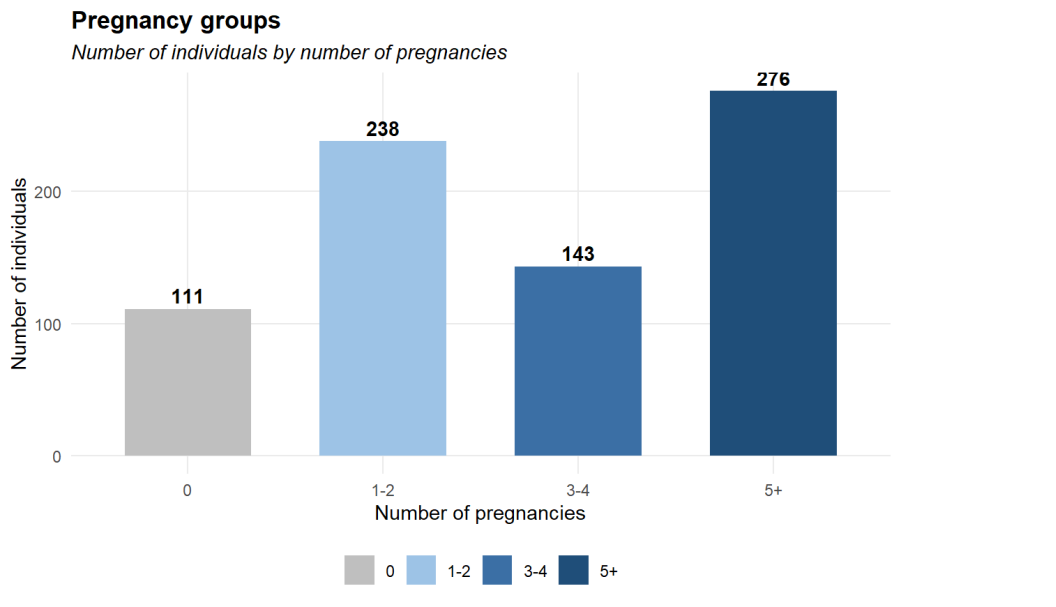


Fig 6. BMI boxplot, outliers up to 67.1

Median BMI: 32.3 · IQR: 27.5–36.6 · Upper outliers retained (up to 67.1)

Sample skewed toward overweight/obese range, relevant to classification model performance

Pregnancy Distribution



Nbr of pregnancies	Full sample (n=768)	Ever-pregnant only (n=657)
Mean	3.8	4.49
Median	3	4
Mode	1	1
Min	0	1
Max	17	17
Range	17	16
SD	3.4	3.22
IQR	(1-6) 5	-

Fig 7. Pregnancy groups distribution

Pregnancies, diabetic vs non-diabetic: 4.87 vs 3.30 · Wilcoxon W = 50985, p = 3.7×10^{-8}-> significant difference

Correlation Matrix: Continuous Variables

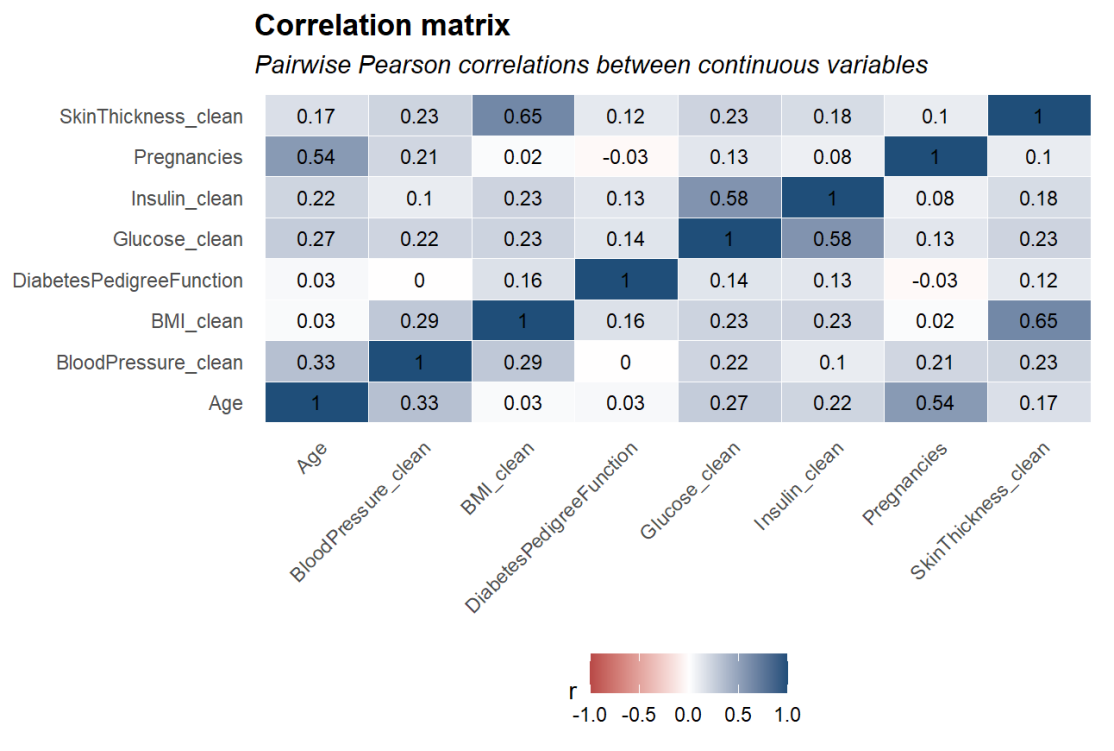


Fig 8. Pairwise Pearson correlations

Key Correlations

BMI ↔ Skin Thickness	**r = 0.65**
Insulin ↔ Glucose	**r = 0.58**
Age ↔ Pregnancies	**r = 0.54**
Age ↔ Blood Pressure	**r = 0.33**
BMI ↔ Blood Pressure	**r = 0.29**

Note: No pair severe enough to destabilise regression, but BMI/SkinThickness (r=0.65) adds coefficient uncertainty

Section 3

Diabetes Rates by Subgroup

Diabetes Rate by BMI Category

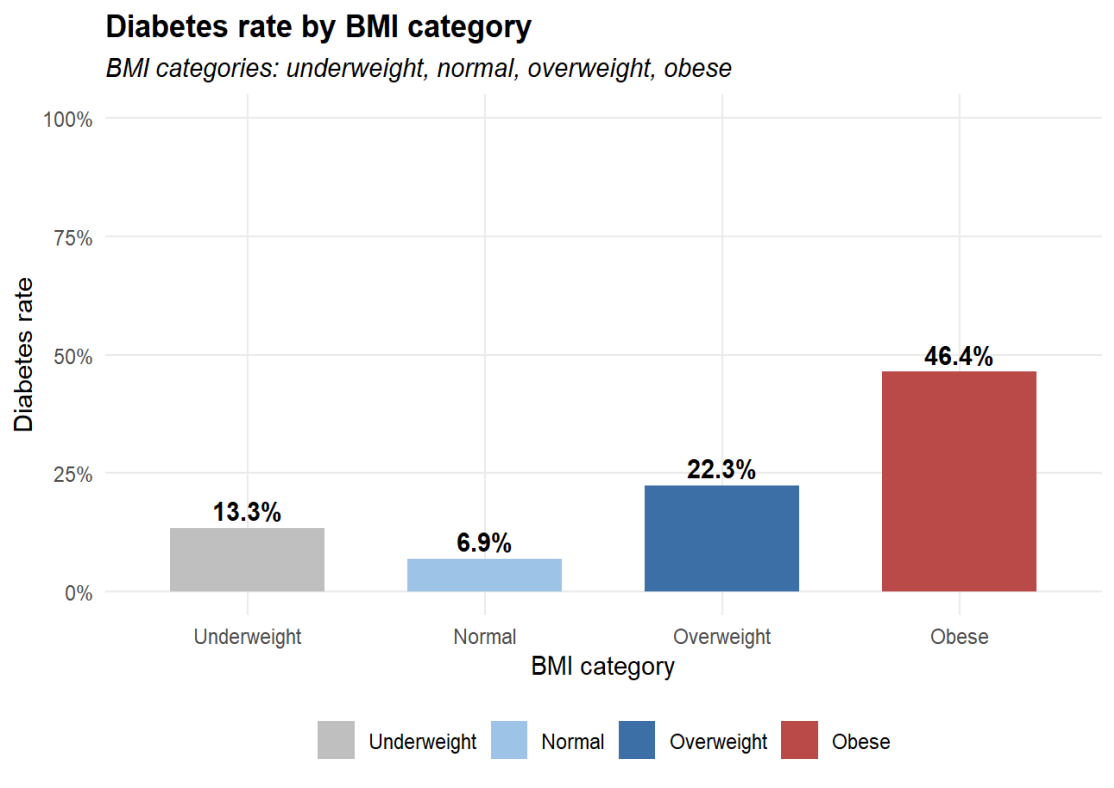


Fig 9. $\chi^2 = 78.25$, $p < 2.2 \times 10^{-16}$ (Pearson chi-square)

Dose-Response Gradient

6.9%
Normal BMI

13.3%
Underweight

22.3%
Overweight

46.4%
Obese

Strong dose-response association supports BMI as an independent predictor of diabetes risk (OR = 1.09, $p < 4.3 \times 10^{-10}$)

Diabetes Rate by Age Group

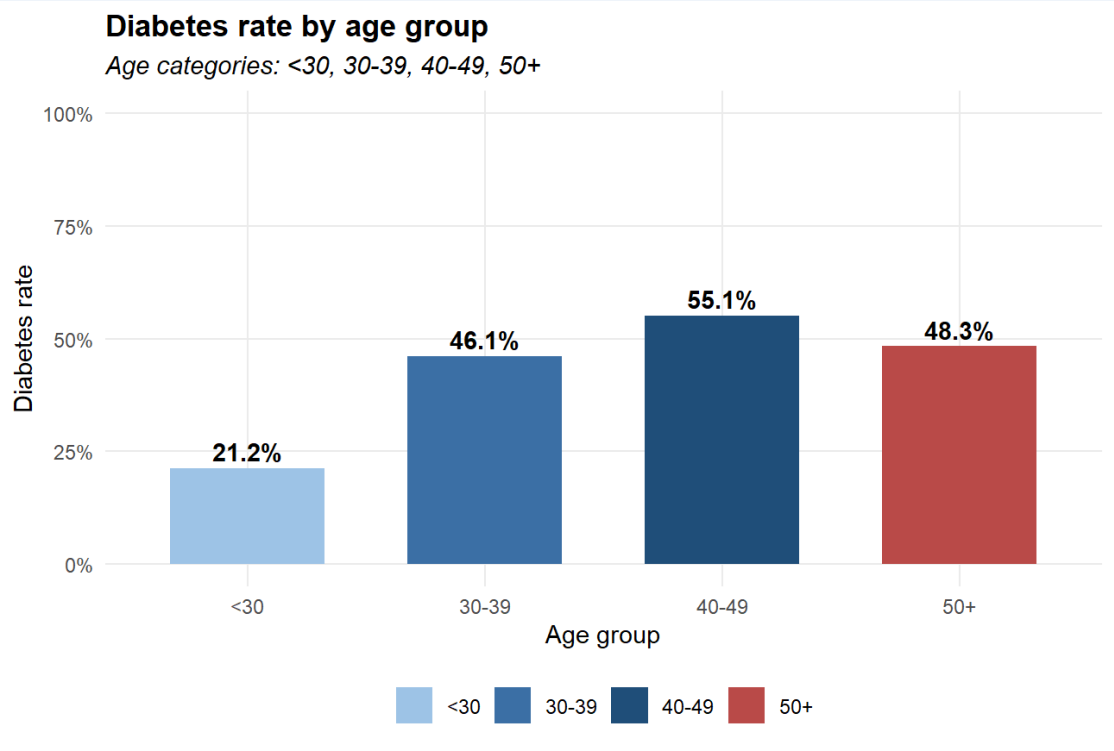


Fig 10. $\chi^2 = 69.92$, $p = 4.5\times10^{-15}$ (Pearson chi-square)

Age as a Risk Escalator

21.2%
Under 30

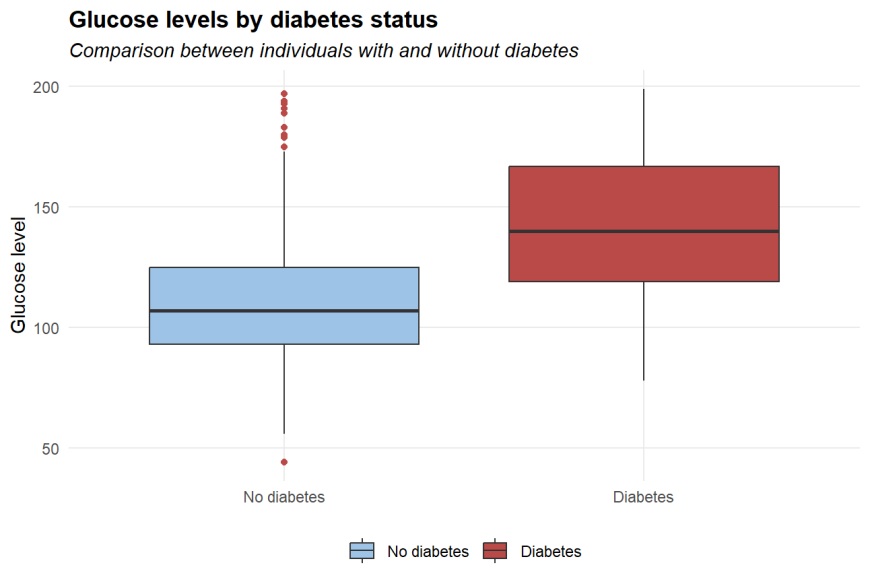
46.1%
30–39

55.1%
40–49

48.3%
50+

⚠ **Plateau between 40–49 and 50+: relationship is not strictly linear across full range**

Glucose Levels by Diabetes Status



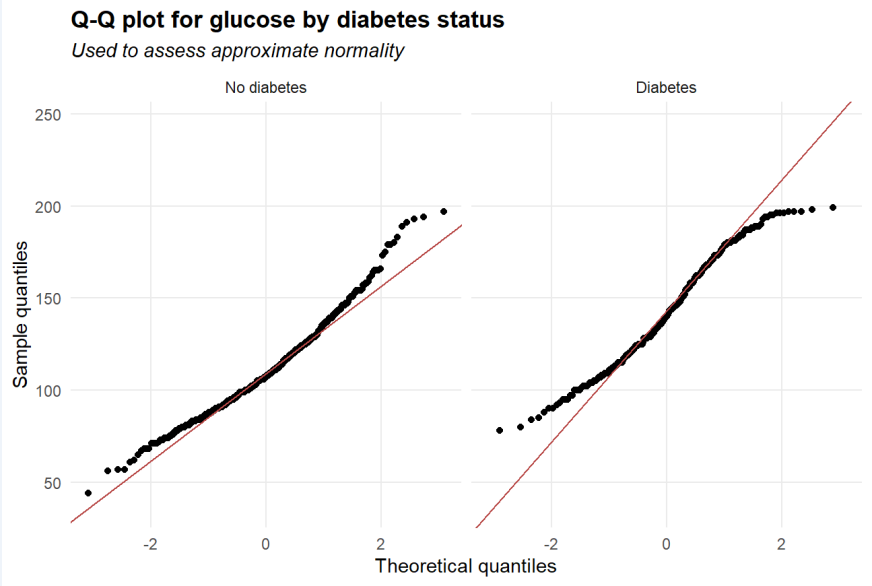


Fig 11. Glucose boxplot by diabetes status

Fig 12. Q-Q plots assessing normality assumption

Diabetic mean: 142.3 mg/dL vs Non-diabetic mean: 110.6 mg/dL

Welch t-test: $t = -14.884$, $p < 2.2\times10^{-16}$ · Wilcoxon rank-sum: $p < 2.2\times10^{-16}$

Diabetes Rate by Number of Pregnancies

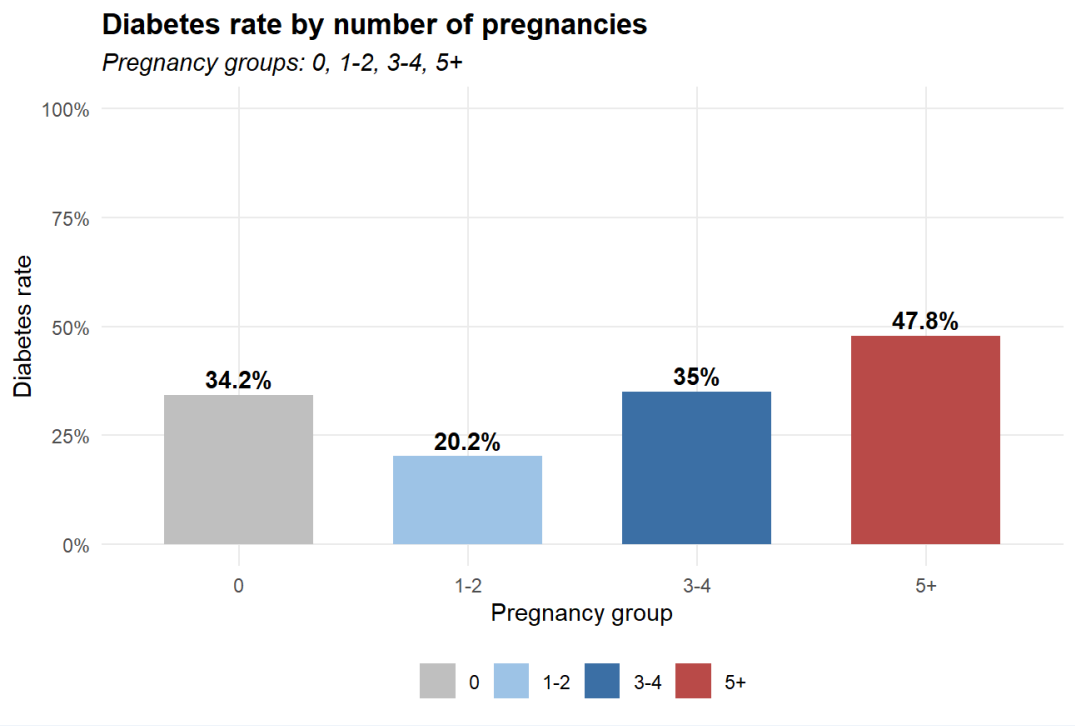


Fig 13. Diabetes rate by pregnancy group

Diabetes Rate by Pregnancy Group

0 → 34.2%
1–2 → 20.2%
3–4 → 35.0%
5+ → 47.8%

Steady rise from 1–2 onward, as expected.
Zero-pregnancy group breaks the pattern
Likely reverse causation: undiagnosed insulin resistance may suppress fertility

Cross-sectional design precludes causal confirmation (Dabelea et al., 2000)

Section 4

Inferential Statistics & Regression Models

Glucose Levels Across Age Groups

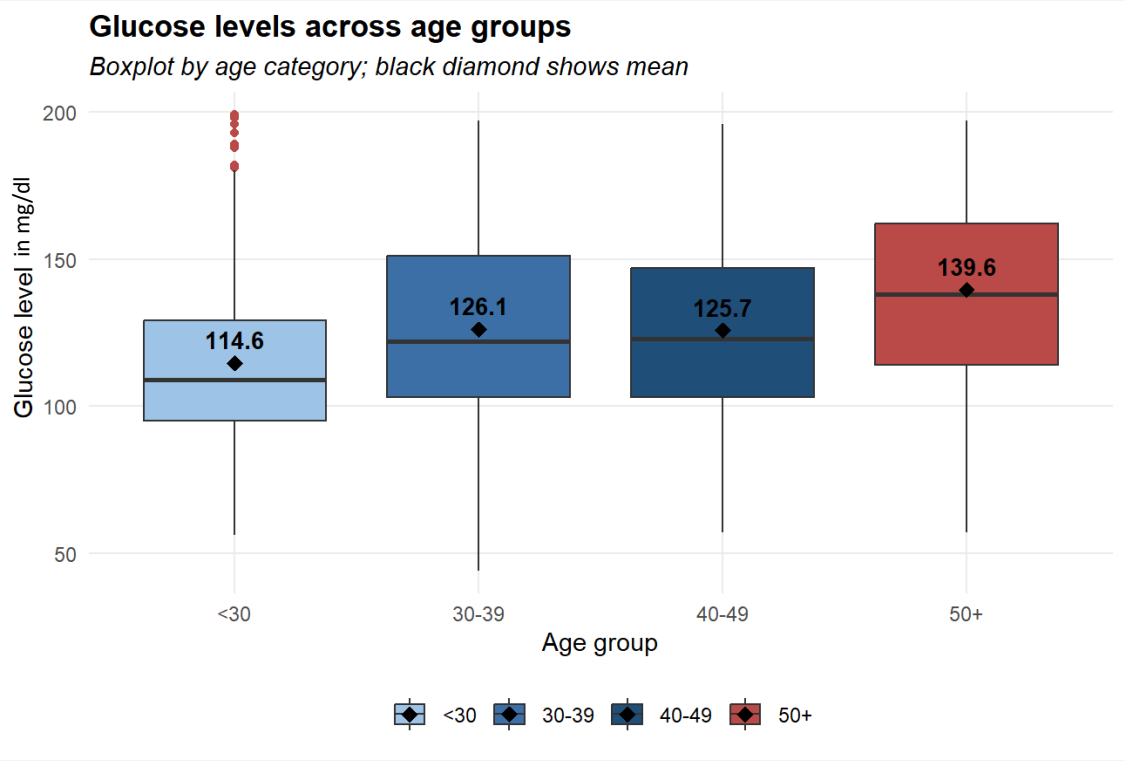


Fig 14. Glucose boxplot by age group

Findings

Mean Glucose by Age Group (mg/dL):

114.6
126.1
125.7
139.6

ANOVA $F(3, 759) = 20.44$, $p < 0.001$
Kruskal-Wallis: $H(3) = 59.04$, $p = 9.4 \times 10^{-13}$
(non-parametric robustness check: upper-tail deviation in <30 group)

Tukey HSD: 30–39 and 40–49 not significantly different from each other

50+ significantly higher than all other groups

<30 group: several high outliers (insulin resistance?)

Multiple Linear Regression: Predicting Glucose

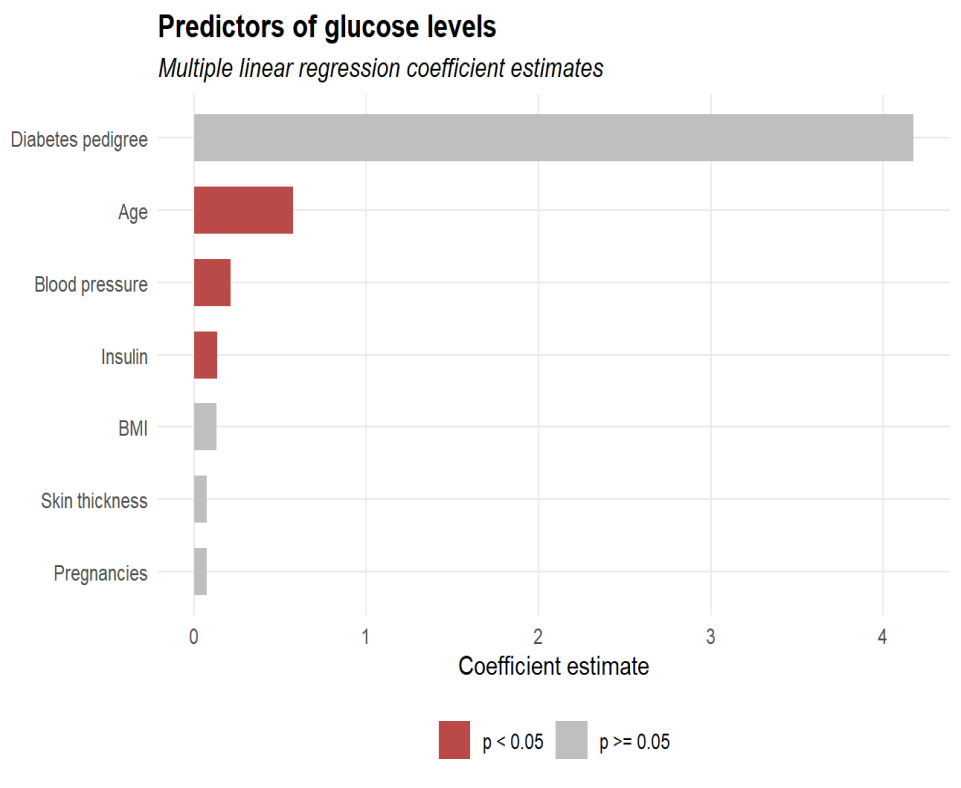


Fig 15. Coefficient estimates (red = p < 0.05)

Model Performance

Adjusted $R^2 = 0.39$

39% of glucose variance explained

$F(7,384) = 36.76$

$p < 2.2 \times 10^{-16}$ overall model significant

Significant Predictors (p < 0.05)

Insulin: strongest physiological predictor
Age: independent life-course effect
Blood pressure - also significant (p < 0.05)

⚠ **Diabetes pedigree:** large raw coefficient (~4.2) but p ≥ 0.05
Scaling artefact, unitless score not comparable to physiological measurements without standardisation

Linear Regression: Diagnostic Plots

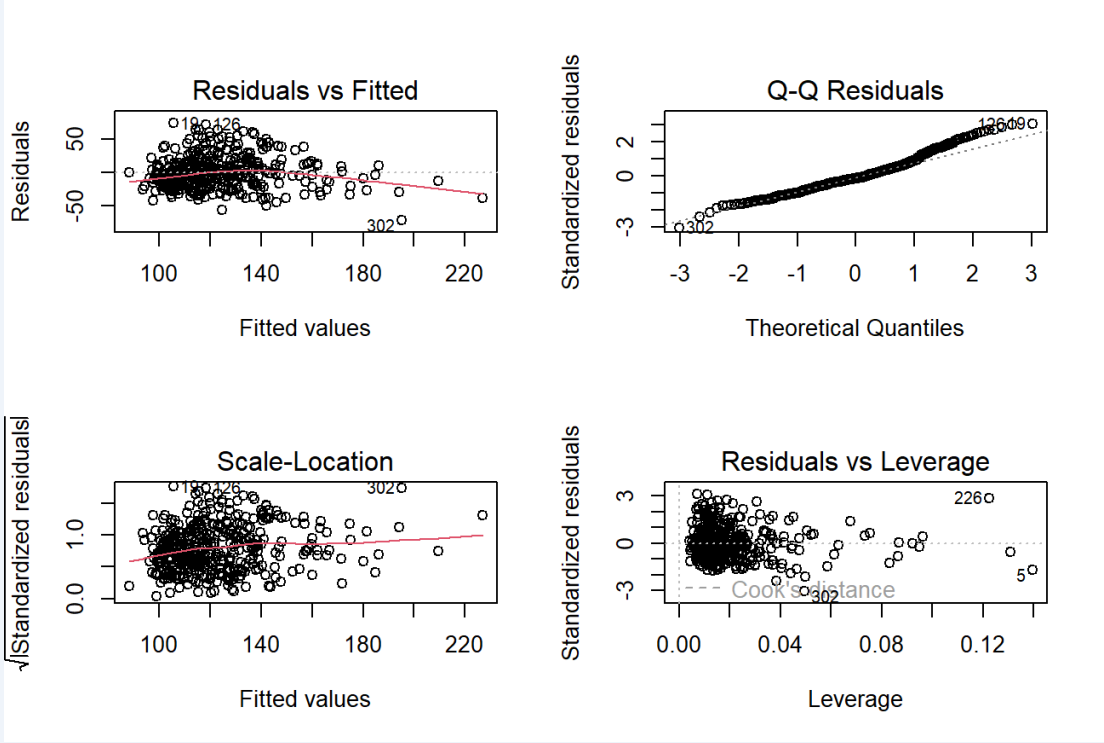


Fig 16. Standard diagnostic plot panel (R base graphics)

Diagnostic Findings

Residuals vs Fitted

Mild systematic pattern, non-linearity possible

Q-Q Residuals

Acceptable normality with upper-tail deviation

Scale-Location

Mild upward trend → non-constant error variance at high fitted values

Leverage

Obs. 226 & 5 have elevated leverage, no Cook's Distance breach

Linear Regression: Actual vs Predicted Glucose

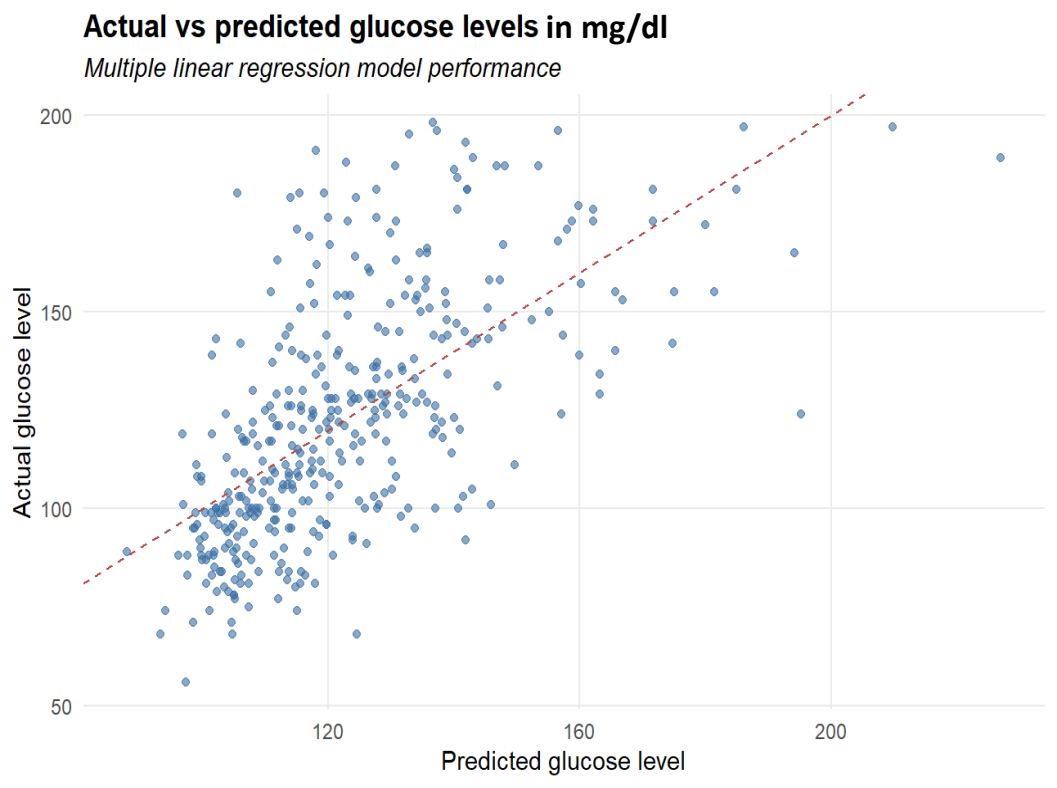


Fig 17. Predicted vs actual glucose (dashed = perfect prediction)

Model Interpretation

Purpose:
Explain glucose mechanism, not clinical prediction

Wide scatter
60% of variance unaccounted for

Missing predictors likely include:
- Diet & physical activity
- HbA1c & fasting status
- Diabetes duration

Section 5

Logistic Regression & Model Evaluation

Logistic Regression: Three-Predictor Model (BMI, Age, Glucose)

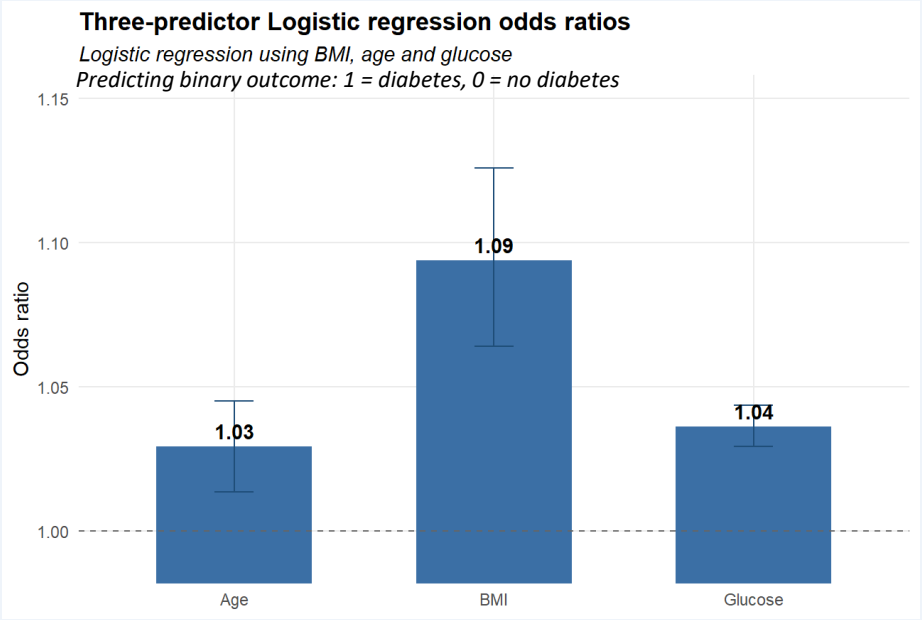


Fig 18. Three-predictor odds ratios

Odds Ratio Interpretation

BMI — **OR = 1.09**
95% CI: 1.06–1.13
Each unit ↑ = 9% higher odds of diabetes
$p < 4.3 \times 10^{-10}$

Glucose — **OR = 1.04**
95% CI: 1.03–1.04
Each mg/dL ↑ = 4% higher odds
$p < 2 \times 10^{-16}$

Age — **OR = 1.03**
95% CI: 1.01–1.05
Each year ↑ = 3% higher odds, significant in this model
p = 0.000238

Model 1: Logistic Regression: Four-Predictor Model (BMI, Age, Glucose, Pregnancies)

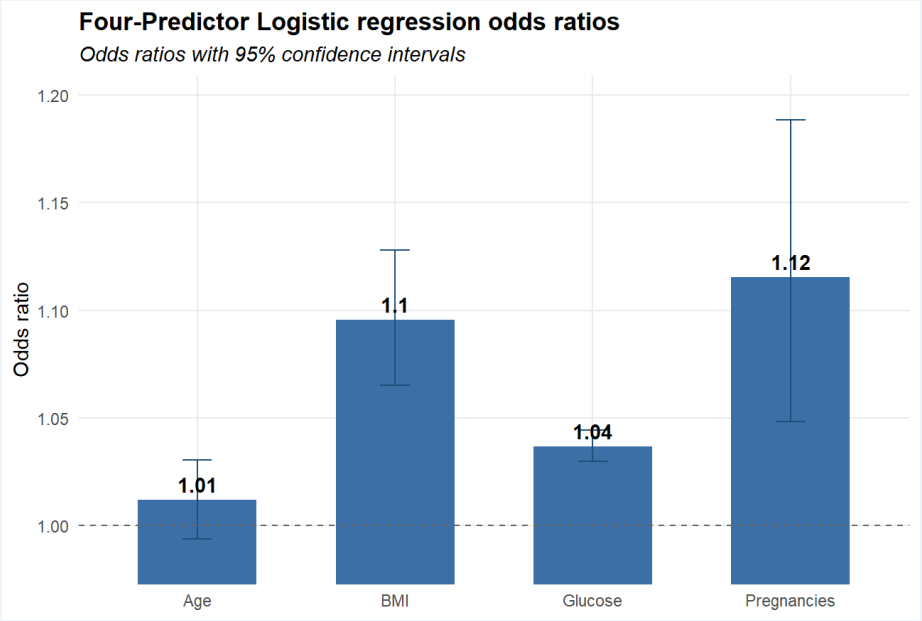


Fig 19. Odds ratios with 95% confidence intervals

Odds Ratio Interpretation

BMI — **OR = 1.10**
95% CI: 1.07–1.13
Each unit ↑ = 10% higher odds of diabetes
$p < 4.6 \times 10^{-10}$

Glucose — **OR = 1.04**
95% CI: 1.03–1.04
Each mg/dL ↑ = 4% higher odds
$p < 2 \times 10^{-16}$

Age — **OR = 1.01**
95% CI: 0.99–1.03
Each year ↑ = 1% higher odds, not statistically significant
p = 0.194

Pregnancies — **OR = 1.12**
95% CI: 1.05–1.19
Each additional pregnancy = 12% higher odds
p < 0.001

Glucose and BMI: Joint Relationship by Diabetes Status

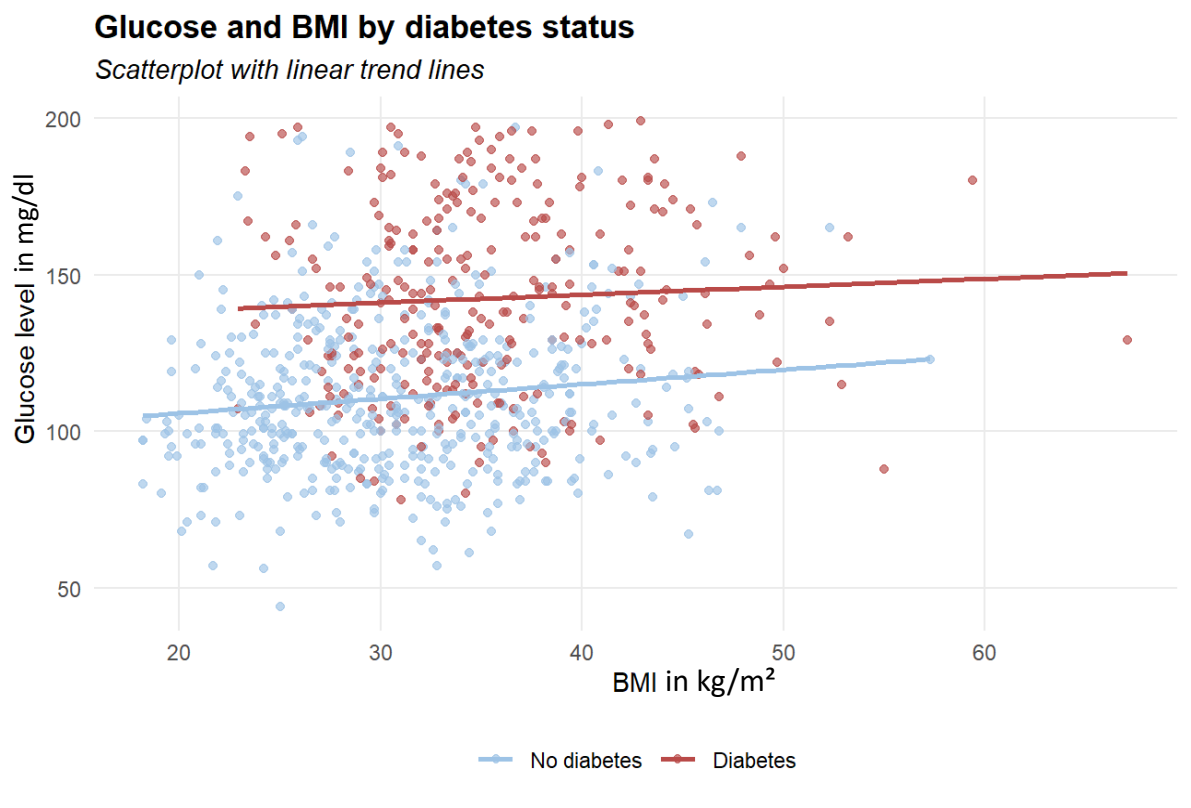


Fig 20. Scatterplot with linear trend lines by diagnosis status

Key Observation

Same BMI → diabetics sit higher on glucose

Glucose & BMI carry **non-redundant information**

Both should be considered in diabetes risk screening models.

(Supports Recommendation 1)

Model 2: Logistic Regression: BMI × Age Interaction

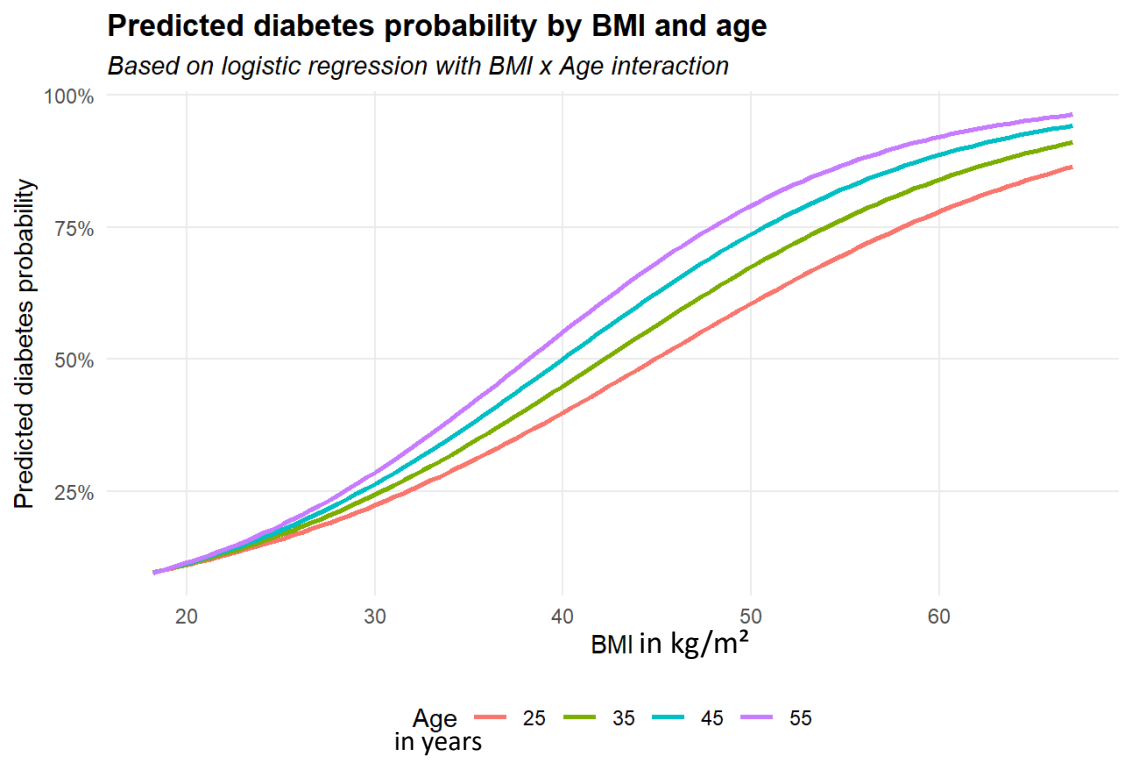


Fig 21. Logistic regression with BMI × Age interaction (van de Schoot et al., 2021)

Interaction Analysis

LRT Result:
Δdeviance = 0.564, p = 0.453 → NOT significant
Interaction OR = 1.001
95% CI spans 1

Without interaction (Model 1)
AIC = 722.9 ✓ preferred

With interaction (Model 2)
AIC = 724.3

Curves widen visibly at BMI > 35

→ power limitation, not a true null

Logistic Regression: Classification Performance

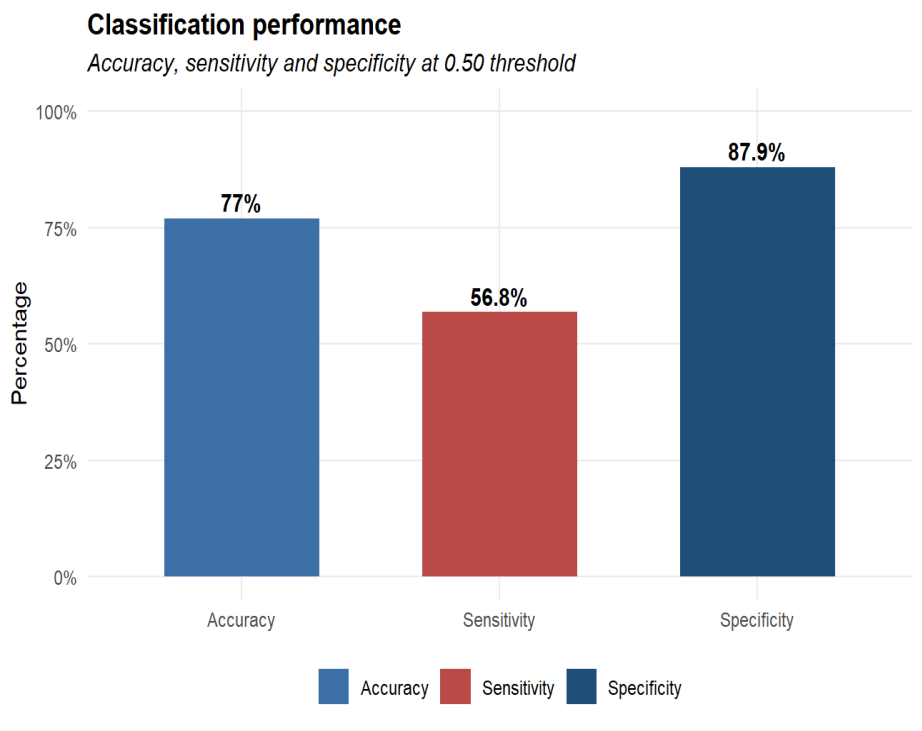


Fig 22. Performance at 0.50 threshold

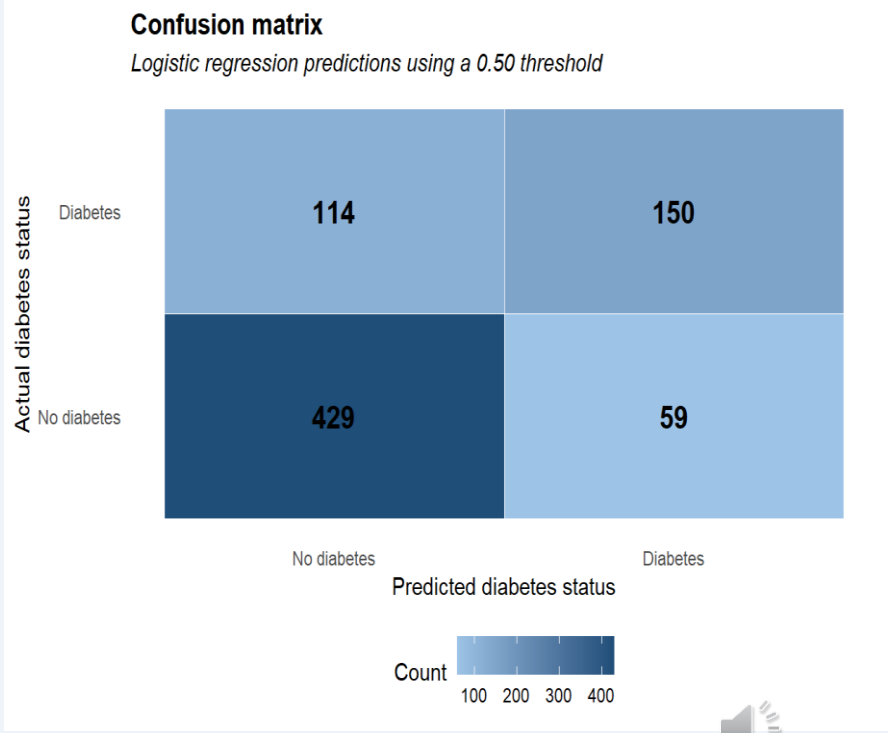


Fig 23. Confusion matrix (logistic regression)

77.0% accuracy · 87.9% specificity · **56.8% sensitivity ⚠** · Hosmer-Lemeshow: p = 0.197 (adequate fit) (Hosmer et al., 2013)

114 true diabetics misclassified as negative vs 59 false positives → 2:1 false negative ratio

Section 6

Conclusions & Recommendations

Summary of Key Findings

Glucose

Mean 142.3 vs 110.6 mg/dL · OR = 1.04 · $p < 2\times10^{-16}$

Insulin

Strongest in linear regression · mechanistic role, not classificatory

BMI

6.9% → 46.4% dose-response · OR = 1.09 · independent of glucose

Pregnancies

34.2% (0 preg.) vs 20.2% (1–2) · likely reverse causation

Age

21.2% (<30 years) → 55.1% (40–49 years, peak) · significance not stable across model

MLR model

Adjusted $R^2 = 0.39$ · 60% variance unexplained

Critical Appraisal: Limitations

Missing Data

768 → 392 · exclusion not confirmed random · selection bias possible

External Validity

No train-test split · in-sample figures likely optimistic

Non-constant error variance

Scale-Location: mild upward trend · SEs underestimated at high glucose

Population Specificity

Pima prevalence ~35% vs ~10% global · recalibration required

Collinearity

BMI–Skin r=0.65 · Insulin–Glucose r=0.58 · coefficient uncertainty understated

Statistical vs Clinical

$p < 0.05 \neq$ clinical fit · 56.8% sensitivity · 0.50 threshold inappropriate

Six Recommendations

1 **Prioritise Glucose & BMI**

Consistent, independent, non-redundant · obligatory screening inputs

2 **Reconsider 0.50 Threshold**

Try 0.30–0.35 · reflect clinical cost of missed diagnosis vs false referral

3 **Age & Pregnancy as Context**

Significant but not modifiable · contextual markers, not causal claims

4 **Multiple Imputation**

Replace complete-case analysis · sensitivity analysis to bound residual bias

5 **External Validation**

No train-test split applied · cross-validation is minimum requirement

6 **Extend to Bayesian Methods**

Informative priors justified · posterior handles class imbalance explicitly

Core Takeaway

Glucose and BMI are the most consistent predictors of diabetes risk. Age is significant in the simpler model, but loses significance once pregnancy history is included

The logistic model is adequate for population-level risk stratification.
Sensitivity limitations, missing data, absence of external validation, and non-constant error variance are conditions, not footnotes.

References

Dabelea, D. *et al.* (2000) 'Intrauterine exposure to diabetes conveys risks for type 2 diabetes and obesity: a study of discordant sibships', *Diabetes*, 49(12), pp. 2208–2211. Available at: https://doi.org/10.2337/diabetes.49.12.2208

Hosmer, D.W., Lemeshow, S. and Sturdivant, R.X. (2013) *Applied Logistic Regression*. 3rd edn. Hoboken: Wiley. Available at: https://doi.org/10.1002/9781118548387

International Diabetes Federation (2021) *IDF Diabetes Atlas*. 10th edn. Brussels: International Diabetes Federation. Available at: https://diabetesatlas.org (Accessed: 19 June 2026).

Khare, A. D. (2022) *Diabetes Dataset* [Dataset]. Kaggle. Available at: https://www.kaggle.com/datasets/akshaydattatraykhare/diabetes-dataset (Accessed: 5 June 2026).

Smith, J.W. *et al.* (1988) 'Using the ADAP learning algorithm to forecast the onset of diabetes mellitus', *Proceedings of the Annual Symposium on Computer Application in Medical Care*. American Medical Informatics Association, pp. 261–265. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC2245318/ (Accessed: 5 June 2026).

Van de Schoot, R. *et al.* (2021) 'Bayesian statistics and modelling', *Nature Reviews Methods Primers*, 1, article 1. Available at: https://doi.org/10.1038/s43586-020-00001-2

World Health Organization (2000) *Obesity: Preventing and Managing the Global Epidemic*. Geneva: World Health Organization. Available at: https://www.who.int/publications/i/item/obesity-preventing-and-managing-the-global-epidemic (Accessed: 25 June 2026).

World Health Organization (2006) *Definition and Diagnosis of Diabetes Mellitus and Intermediate Hyperglycaemia*. Geneva: World Health Organization. Available at: https://www.who.int/publications/i/item/definition-and-diagnosis-of-diabetes-mellitus-and-intermediate-hyperglycaemia (Accessed: 25 June 2026).